

1. Introduction

This project explores the development of Convolutional Neural Networks (CNNs) for age estimation and gender classification using facial images. Two models were developed: Model A, trained from scratch with specific architectural constraints, and Model B, a fine-tuned pre-trained model. The models were evaluated on unseen validation data using Mean Absolute Error (MAE) for age estimation and accuracy for gender classification, ensuring robustness and generalisation.

2. Model A

2.1 Model Architecture and Design

The CNN architecture was carefully designed to handle multi-output predictions, allowing for age and gender classification within a single network. The model was structured with shared convolutional layers, followed by two task-specific branches.

2.2 Shared Convolutional Backbone

The model processes images through a series of convolutional layers to extract meaningful features while progressively reducing spatial dimensions. The 3×3 convolutional filters were used to balance computational efficiency and feature extraction, with the number of filters increasing from 32 to 256 for progressively deeper feature representation.

To adhere to the architectural constraints, the final shared feature map before splitting into task-specific branches was 8×8 , ensuring it remained below the 10×10 limit imposed on the fully connected layers.

2.3 Age Estimation Branch

The age branch directly receives the 8×8 feature map, which is flattened and passed into fully connected layers optimised for regression.

- Batch Normalisation was applied to stabilise learning and reduce internal covariate shift.
- Dropout layers were strategically implemented to prevent overfitting, with 0.25 dropout after convolutional pooling and 0.5 in dense layers for robust generalisation.

Loss Function Selection

Two regression loss functions were compared:

- Mean Squared Error (MSE): Penalises large errors more aggressively but can lead to unstable gradients.
- Mean Absolute Error (MAE): Provided smoother gradients and better predictive stability, making it the final choice.

2.3 Gender Classification Branch

The gender classification branch extends from the shared 8×8 feature map, adding additional convolutional layers to further reduce dimensions to 4×4 .

During initial training, gender classification underperformed, suggesting under-fitting. Additional convolutional layers were added, significantly improving accuracy by refining gender-specific feature representations.

- Binary Cross-Entropy was used as the loss function, as it is well-suited for binary classification.
- A single neuron with Sigmoid activation ensured probabilistic output, suitable for the binary task.

2.4 Data Augmentation Strategy

To enhance generalisation, multiple data augmentation techniques were tested. Initially, complex augmentations such as brightness shifts, zoom, and translations were explored, but these led to degraded model performance due to distortions in essential facial features.

The final strategy focused on horizontal flips and slight rotations, which preserved facial features.

2.5 Optimised Training Procedure

To achieve optimal learning, specific training strategies were implemented:

- Optimiser: Adam, selected for adaptive learning rates to ensure efficient training and convergence.
- Epochs: 100, allowing the model to sufficiently learn without premature termination.
- EarlyStopping: Set with a patience of 20 epochs and a minimum delta of 0.001, ensuring training stopped at the best generalisation point.
- ReduceLROnPlateau: Implemented with a patience of 3 epochs and a delta of 0.005, reducing the learning rate upon stagnation to prevent plateaus and maintain training momentum.

2.6 Learning Curve Insights

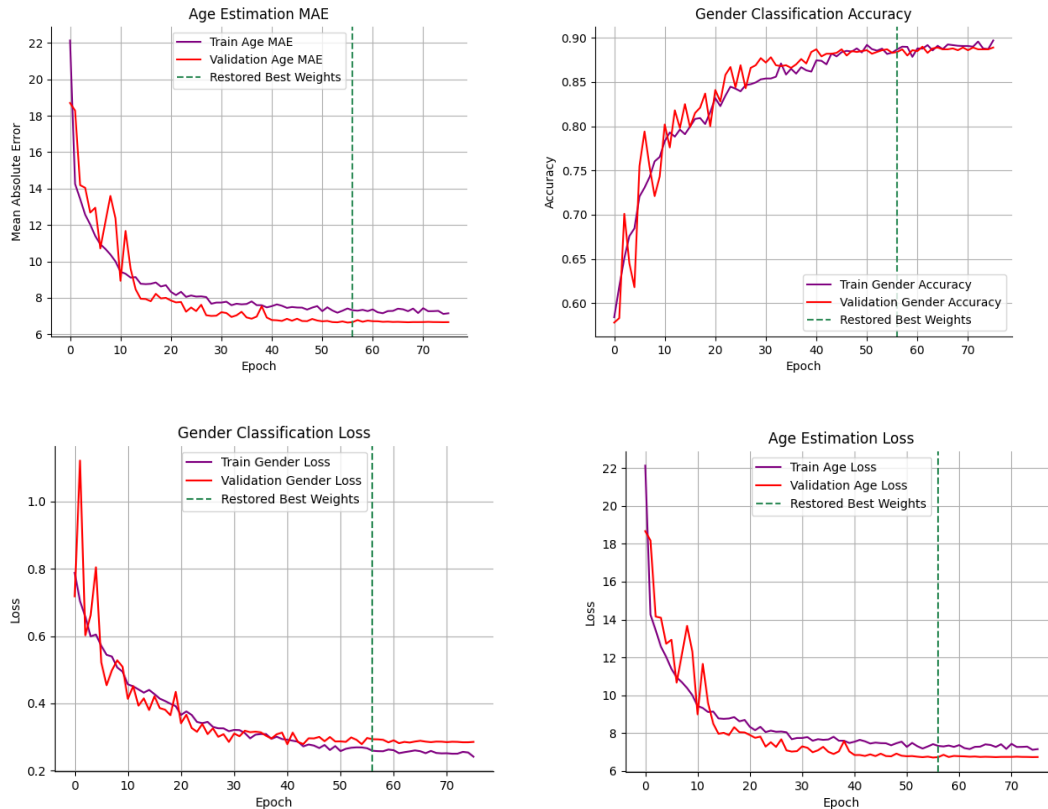


Figure 1: The learning curves for Model A.

The learning curves indicate stable training and minimal overfitting, showcasing strong generalisation across both tasks.

Gender Classification Performance

- Accuracy increased rapidly, stabilising at ~88% on validation data, closely matching training accuracy.
- Training and validation loss curves followed a smooth downward trend, indicating effective convergence and minimal overfitting.

Age Estimation Performance

- MAE consistently decreased, reaching a final validation MAE of ~6.4 years, aligning closely with training MAE.
- The loss curve displayed smooth, stable convergence, confirming the model's robustness and reliability.

Best Weights Selection and Stability

To prevent overfitting, the model's best weights were selected using EarlyStopping and Restore Best Weights. The green vertical dashed line on the learning curves indicates the epoch where validation performance peaked, ensuring that the final model retained optimal generalisation.

2.7 Final Model Performance and Generalisation

The CNN model demonstrated strong performance on unseen test data, achieving:

- Near-zero overfitting, confirmed by aligned training-validation curves.
- Robust generalisation, ensuring stability in real-world scenarios.
- Well-tuned hyper-parameters, balancing computational efficiency and predictive power.

3. The Pre-Trained CNN

3.1 Overview of the Pre-Trained Model

The convolutional neural network (CNN) utilised as the pre-trained CNN was VGG16, a deep learning model trained on 14 million labeled images across thousands of categories (Markoff, 2012). It is a 16-layer CNN, consisting of 13 convolutional layers and 3 fully connected layers (GeeksforGeeks, 2020). By leveraging its already learned generic features, VGG16 was applied to the significantly smaller UTKFace training dataset using the principles of transfer learning.

3.2 Model B Architecture

The fully connected layers of VGG16 were removed, as custom classification layers were defined. Initially, the pre-trained convolutional layers were frozen. This allowed the model to leverage its learned feature representations from ImageNet without modifying them.

The feature map resulting from the VGG16 base layers was flattened into a one-dimensional vector that was then fed into a fully connected layer with 1024 neurons and ReLU activation. The number of neurons in this layer was manually tuned.

For the output layers, a fully connected layer with a single neuron and a linear activation function was used for age prediction, as age is a continuous variable. A separate fully connected layer with a single neuron and a sigmoid activation function was employed for gender classification.

3.3 Training the Model

The model was compiled with an initial learning rate of 10^{-3} , using the same optimiser, metrics, and loss functions as Model A. A batch size of 32 was chosen after manually tuning the hyperparameter, as it offered a good balance between computational efficiency and training stability.

Training was set for a maximum of 100 epochs, but with early stopping enabled to avoid unnecessary computation. If the validation loss failed to improve for three consecutive epochs, the learning rate was reduced by a factor of half. If the validation loss continued to plateau for another two epochs (a total of five consecutive epochs), training was stopped early, and the best weights obtained during the training process were restored. This approach ensured that we avoided unnecessary overfitting.

3.4 Fine-Tuning the VGG16 Model

Following initial training, the last 20 layers of VGG16 were unfrozen to allow fine-tuning. This step enabled the model to adjust high-level feature representations to better align with the current classification task. The model was then recompiled with a lower learning rate (10^{-5})

and additional training was performed for up to 100 epochs, using the updated learning rate and unfrozen layers.

3.5 Transfer Learning Approach

Transfer learning enables the application of a pre-trained model (VGG16), which has learned features from a large dataset (ImageNet), to a new, smaller dataset (4000 images from the UTKFace dataset). This approach provided several advantages, including a reduction in training time, as lower-level features did not need to be re-learned, allowing the model to focus on refining more complex patterns.

3.6 Evaluation of model performance

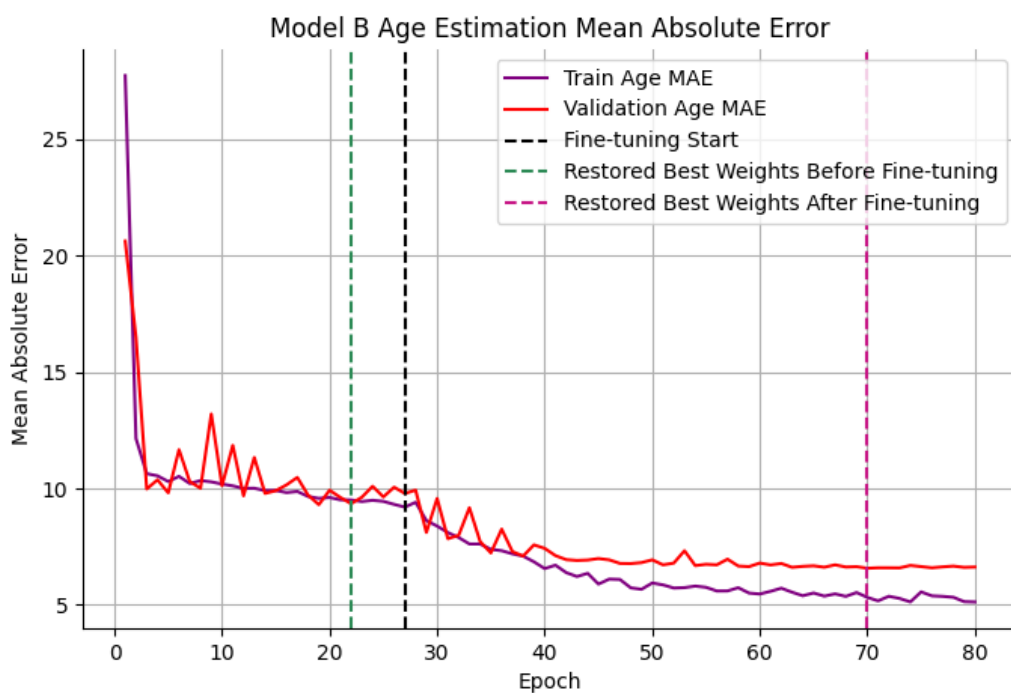


Figure 2: The mean absolute error on age, Model B.

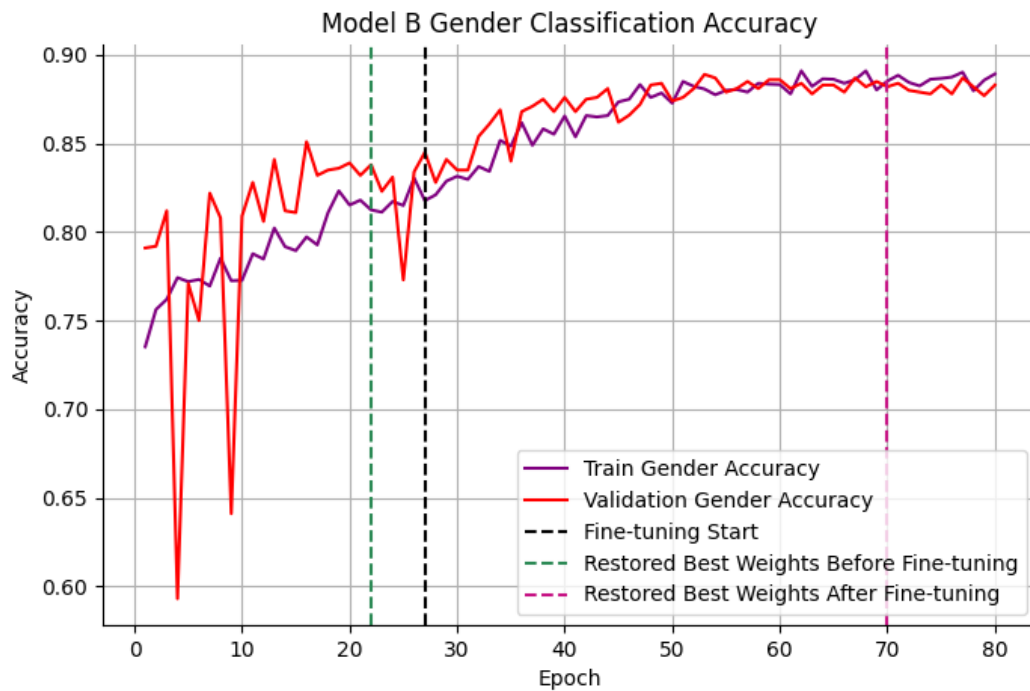


Figure 3: The gender classification accuracy, Model B.

In the age estimation task, both training and validation MAE decrease rapidly in the initial epochs, but the validation MAE exhibits fluctuations, indicating potential instability in generalisation. Fine-tuning, introduced at epoch 27 further improves the validation age MAE before it eventually plateaus at approximately 6.6. The best weights for the model were restored from epoch 70, disregarding any overfitting that occurred after this. The training age MAE is approximately one year below the validation age MAE, pointing to no significant overfitting, suggesting mild overfitting.

By the final epochs, both training and validation gender accuracy converge around 88%, indicating strong generalisation. No overfitting is observed.

4. Summary

Both models performed similarly in both age estimation and gender classification, with the restored best-weight validation gender accuracy for both models reaching approximately 88% and the validation age MAE stabilising around 6.6. However, Model A achieved its optimal performance in 56 epochs, whereas Model B required 70 epochs, indicating that Model A was more efficient. This is further reflected in training time, where Model A completed in 25 minutes compared to Model B's 30 minutes.

Additionally, Model A exhibited a smoother and more stable improvement in validation, while Model B showed more fluctuations in the earlier epochs before stabilising post fine-tuning, suggesting Model A was better able to generalise to unseen images. Given these factors, Model A appears to be the more efficient choice, achieving comparable results in fewer epochs and with a shorter runtime.

A limitation of both models is that they were trained on only cropped and aligned faces, so their performance may be worse on images of faces that are partially occluded (e.g., by glasses or masks). Training on a more varied dataset with unprocessed images could help improve the robustness of both models.

The models' performance could also be improved through hyperparameter tuning, using techniques like random search or grid search to systematically explore different learning rates, batch sizes, and dropout rates. The development of a custom optimiser could potentially reduce training time, but would require extensive tuning.

Age and gender classification models have practical applications in areas such as facial recognition on smartphones and targeted advertising. They can also be integrated into security systems, particularly in crowded environments like airports, malls, or stadiums. However, deploying these models in real-world scenarios raises ethical concerns regarding privacy and bias.

5. References

GeeksforGeeks (2020). VGG-16 | CNN model. [online] GeeksforGeeks. Available at: <https://www.geeksforgeeks.org/vgg-16-cnn-model/>.

Markoff, J. (2012). Seeking a Better Way to Find Web Images. The New York Times. [online] 19 Nov. Available at: <https://www.nytimes.com/2012/11/20/science/for-web-images-creating-new-technology-to-seek-and-find.html> [Accessed 8 Mar. 2025].